

Tide Gauge

A tide gauge is an oceanographic instrument used to measure changes in sea level over time. It records the rise and fall of seawater caused mainly by the gravitational forces of the Moon and the Sun, which create tides. Tide gauges are installed along coastlines, ports, harbors, estuaries, and islands to continuously monitor sea-level variations. These instruments are very important in oceanography, navigation, climate studies, coastal engineering, and disaster management.

A tide gauge works by measuring the height of seawater relative to a fixed reference point called a vertical datum. In traditional tide gauges, a floating device placed inside a stilling well rises and falls with the water level and records tidal changes. Modern tide gauges use advanced technologies such as pressure sensors, acoustic sensors, ultrasonic sensors, and radar systems to measure sea level more accurately. The collected data is transmitted to computers and monitoring centers for analysis and forecasting.

Tide gauges help scientists study tidal patterns, mean sea level, sea-level rise, storm surges, and tsunami activity. They are also useful for safe navigation because ships depend on accurate tidal information while entering or leaving ports and shallow coastal areas. Long-term tide gauge records provide valuable information about climate change and the effects of global warming on oceans and coastal regions.

Modern tide gauges are often connected to satellite and wireless communication systems that provide real-time sea-level data. These systems are part of global ocean observation networks used by scientific organizations and governments worldwide. Although tide gauges are highly useful, they require regular maintenance because storms, corrosion, and coastal environmental changes can affect their operation and accuracy.

Tide Measurement Units

Sea level is usually measured in: Meters(m), Centimeters(cm)

Working of Tide Gauge

1. Sea water enters the stilling well.
2. Sensor measures water level.
3. Data is recorded continuously.
4. Information is transmitted to monitoring centers.

Types of Tide Gauge :

Tide gauges are instruments used to measure sea level changes, and they are classified into different types based on the technology used. The four main types are float type, pressure sensor type, radar type, and acoustic type.

1. Float Type Tide Gauge (Mechanical Type)

The float type tide gauge is one of the oldest and most widely used methods. It works using a float placed inside a stilling well, which is connected to the sea through small openings. The stilling well reduces the effect of waves, so only the actual rise and fall of sea level is measured. As the sea level changes, the float moves up and down. This movement is transferred through a pulley system or digital encoder to a recording device, which draws a continuous tide graph. This type is very reliable and gives accurate long-term records, but it requires regular maintenance.

2. Pressure Sensor Tide Gauge

The pressure sensor tide gauge works on the principle of water pressure measurement. It is placed underwater at a fixed depth near the seabed. Since water pressure increases with depth, any change in sea level changes the pressure acting on the sensor. The sensor detects this change and converts it into electrical signals, which are then processed to calculate sea level height. This type is useful in deep water and harsh environments, and it is less affected by surface waves. However, it may need calibration due to changes in temperature and water density.

3. Radar (Microwave) Tide Gauge

The radar tide gauge is a modern and highly accurate system. It is installed above the water surface, usually on a bridge or platform. It sends microwave radar signals toward the sea surface, and these signals reflect back to the sensor. The time taken for the signal to return is used to calculate the exact distance between the sensor and the water surface, which gives the sea level. Since it is a non-contact method, it is very safe, requires low maintenance, and works well even in rough sea conditions.

4. Acoustic (Ultrasonic) Tide Gauge

The acoustic tide gauge uses sound waves instead of light or mechanical parts. A sensor placed above the water sends ultrasonic pulses toward the sea surface. These pulses travel through air, hit the water surface, and reflect back to the sensor. The system measures the time taken for the echo to return and calculates the distance to the water surface, which gives the tide level. This method is simple and cost-effective, but it can be slightly affected by air temperature, humidity, and wind conditions.

Tide Gauge Sea Level

Tide gauge sea level refers to the measurement of sea level changes using a tide gauge instrument. A tide gauge measures the height of seawater relative to a fixed reference point called a vertical datum. These measurements help scientists observe the rise and fall of tides, changes in mean sea level, storm surges, and long-term sea-level trends. Tide gauges are usually installed along coastlines, ports, harbors, estuaries, and islands to continuously monitor sealevel variations.

Traditionally, tide gauges used float systems placed inside stilling wells, where the float moved up and down with changing water levels. Modern tide gauges now use advanced technologies such as pressure sensors, acoustic sensors, ultrasonic sensors, and radar systems for more accurate and real-time sea-level measurements. The collected data is transmitted to computers and research centers for monitoring and analysis.

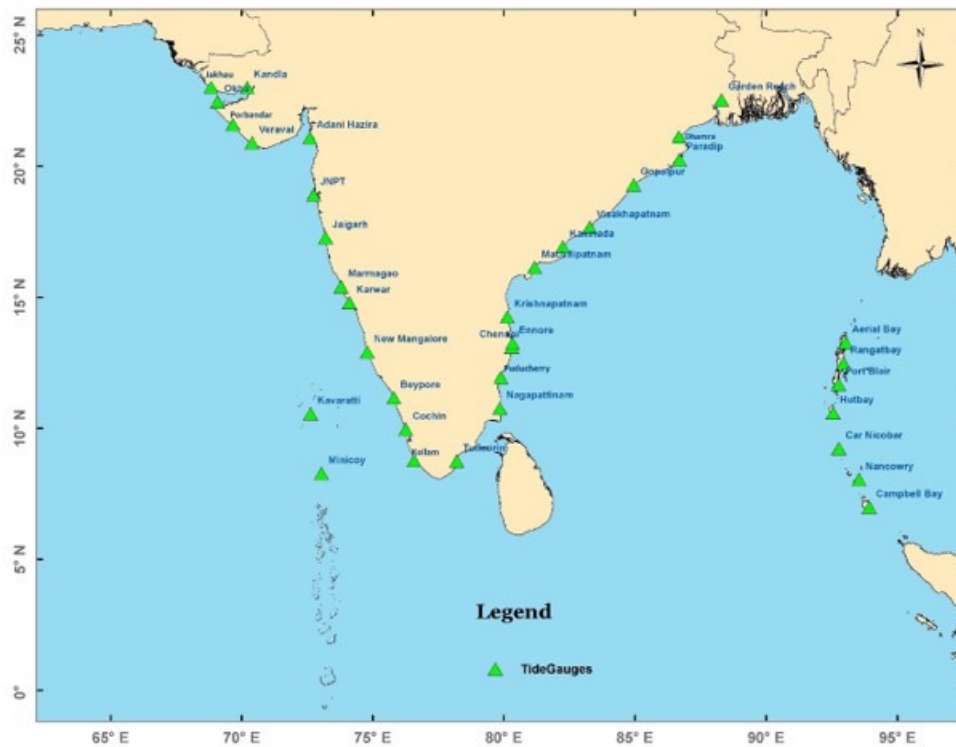
Tide gauge sea-level measurements are important in oceanography, navigation, coastal engineering, and climate studies. They help scientists calculate mean sea level and understand the effects of climate change, melting glaciers, and thermal expansion of seawater. Long-term tide gauge records have shown that global sea level has risen significantly over the past century due to global warming. Tide gauge data is also used in tsunami warning systems because sudden changes in sea level can indicate tsunami activity.

In addition, tide gauge measurements support safe navigation for ships by providing accurate tidal information for ports and coastal waters. Many countries maintain national and international tide gauge networks to continuously monitor sea-level changes and improve disaster management systems. Today, tide gauge observations are often combined with satellite data to improve the accuracy of global sea-level monitoring and climate research.

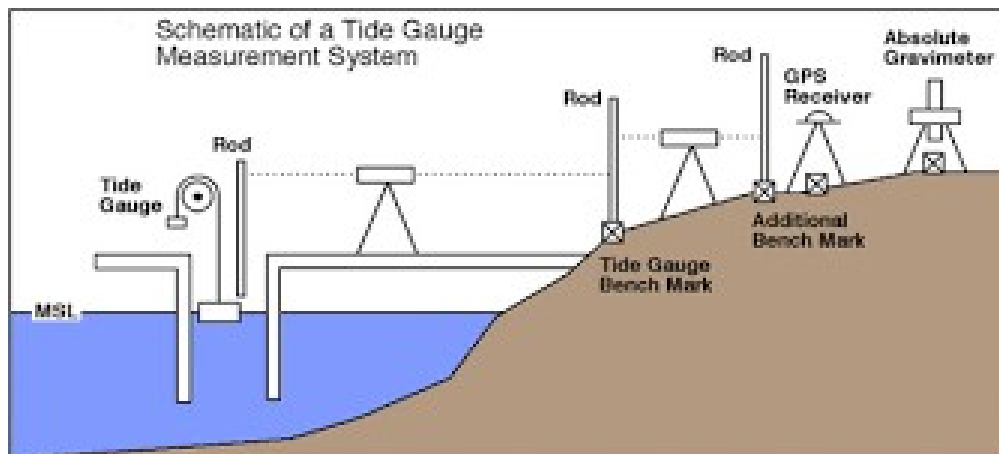
Principles of Tide Gauge

A tide gauge works on the basic principle of measuring the continuous variation of sea level with respect to a fixed reference point (datum) on land. Since land is considered stable compared to the ocean surface, the difference between the sea surface and this reference point gives the tide level. The instrument records how the sea surface moves up and down due to tides, waves, wind effects, atmospheric pressure changes, and long-term sea level variations. This measurement is done either mechanically (float system) or electronically (pressure/radar sensors). In modern systems, sensors detect the height of water and convert it into electrical signals. These signals are stored or transmitted to observation centers, where they are used for studying tides, climate change, and coastal hazards.

Indian Tide Gauges stations established by INCOIS



Schematic of a Tide Gauge measurement system



Tide Gauge Estimates of Mean Sea Level Rise

Estimates of global sea level rise which were derived from tide gauge records are found in the table below. Most of the investigators reported that the estimated values were sensitive to the choice of record length and the tide gauges selected. This sensitivity coupled with different computational techniques and modeling would certainly explain some of the differences shown below.

Error (mm/yr) Data Used (years) # of Tide Gauges References

Sea Level Rise (mm/yr)

Sea Level Rise from Tide Gauge Studies

Sea Level Rise (mm/yr)	Error (mm/yr)	Data Used (years)	# of Tide Gauges	References
2.8	±0.8	1993–2009	~200	Church & White (2011)
1.7	±0.2	1900–2009	>38 since 1900	Church & White (2011)
1.9	±0.4	1961–2009	>190 since 1960	Church & White (2011)
1.43	±0.14	1881–1980	152	Barnett (1984)
2.27	±0.23	1930–1980	152	Barnett (1984)
1.2	±0.3	1880–1982	130	Gornitz and Lebedeff (1987)
2.4	±0.9	1920–1970	40	Peltier and Tushingham (1989)
1.75	±0.13	1900–1979	84	Trupin and Wahr (1990)
1.7	±0.5	N/A	N/A	Nakiboglu and Lambeck (1991)
1.8	±0.1	1880–1980	21	Douglas (1991)
1.62	±0.38	1807–1988	213	Unal and Ghil (1995)

Advantages

Tide gauges provide continuous and long-term data about sea level changes, which is very important for studying oceans and climate. They help scientists understand tides, storm surges, and sea level rise. They are also useful for navigation in ports and harbors, ensuring safe movement of ships. Modern tide gauges are highly accurate and can record data automatically without human intervention. Some types, like radar and pressure sensors, can work in harsh marine conditions and provide real-time information.

1. Provides continuous sea level data (24/7)
2. Helps in predicting tides and coastal flooding
3. Useful for climate change and sea level rise studies
4. Supports safe navigation for ships and boats
5. High accuracy in modern systems
6. Automatic data recording and real-time monitoring
7. Works in most coastal environments

Disadvantages

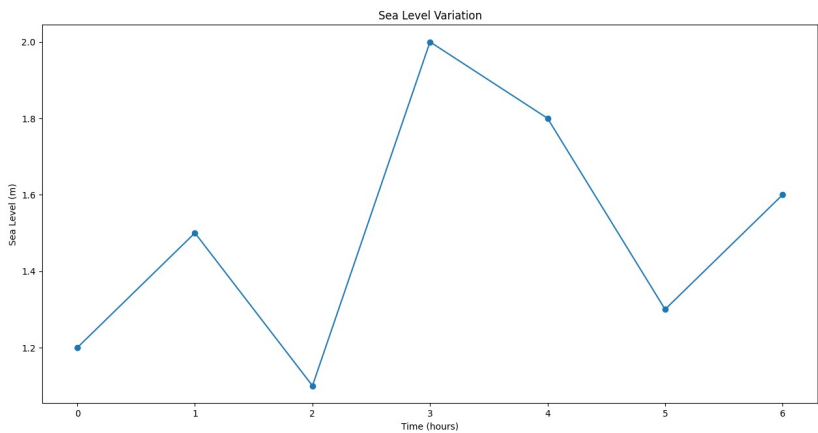
Despite their usefulness, tide gauges also have some limitations. They only measure sea level at specific locations, so they cannot represent the entire ocean. Some types, like float and acoustic gauges, can be affected by waves, wind, and environmental conditions. Installation and maintenance can be expensive, especially for advanced systems. Data accuracy may also be affected by local land movement, such as sinking or uplift of the ground. Additionally, some gauges require regular calibration to maintain accuracy.

1. Measures sea level only at fixed locations
2. Cannot represent full ocean conditions
3. Some types affected by waves and weather conditions
4. Installation and maintenance can be costly
5. Land movement can affect accuracy (uplift or sinking)
6. Some systems require frequent calibration
7. Limited coverage compared to satellite monitoring

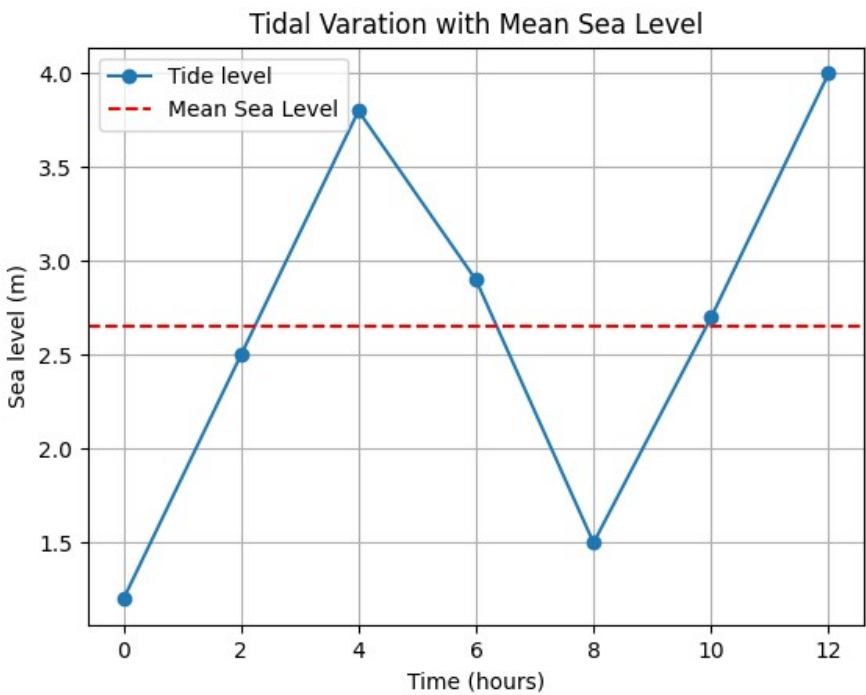
Importance of Tide Gauges in India

Tide gauges play a very important role in India because the country has a long coastline of about 7,500 km, and many people live in coastal areas. These instruments help monitor sea level changes and protect coastal regions from natural hazards. In India, tide gauges are used by organizations like the Indian National Centre for Ocean Information Services (INCOIS) and other coastal research stations to study tides, tsunamis, and climate change impacts. They are especially important for early warning systems. Tide gauges help detect unusual sea level rise that may indicate tsunamis or storm surges, giving authorities time to warn coastal communities. They are also useful for navigation in major ports like Mumbai, Chennai, and Visakhapatnam, ensuring safe movement of ships. In addition, long-term tide gauge data helps scientists understand sea level rise due to global warming, coastal erosion, and changes in monsoon patterns.

Sea Level Variation Analysis



Analysis of Sea Level Variation Over 12-Hour Period



Conclusion

Tide Gauges are important oceanographic instruments used to measure changes in sea level over time. They play a major role in ocean observation, climate change studies, tsunami warning systems, navigation, and coastal management. Data collected from tide gauges helps scientists understand long-term sea level rise, tidal patterns, and the impact of global warming on oceans. Therefore, tide gauges are essential tools for monitoring and protecting coastal environments and supporting oceanographic research.

References

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